
Physics 523 HW 8

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I use the notation in my Real Scalar Field Notes. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the dot product of the momentum 4-vector: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.

Part I

QED

1. LAGRANGIAN

The bare Lagrangian is

$$\mathcal{L} = \bar{\psi}_B(i\cancel{D} - m_B)\psi_B - \frac{F_B^2}{4} - e_B \bar{\psi}_B \cancel{A}_B \psi_B, \quad F^2 = F^{\alpha\beta} F_{\alpha\beta}. \quad (1)$$

I renormalize.

$$m_B = Z_m m, \quad e_B = Z_e e, \quad \psi_B = \sqrt{Z_2} \psi, \quad \vec{A}_B = \sqrt{Z_3} \vec{A} \quad (2)$$

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha \implies F_B^2 = Z_3 F^2 \quad (3)$$

$$\mathcal{L} = Z_2 \bar{\psi}(i\cancel{D} - Z_m m)\psi - Z_3 \frac{F^2}{4} - Z_e Z_2 \sqrt{Z_3} e \bar{\psi} \cancel{A} \psi \quad (4)$$

$$\mathcal{L} = i Z_2 \bar{\psi} \cancel{D} \psi - Z_m Z_2 m \bar{\psi} \psi - Z_3 \frac{F^2}{4} - Z_e Z_2 \sqrt{Z_3} e \bar{\psi} \cancel{A} \psi \quad (5)$$

$$Z_1 = Z_e Z_2 \sqrt{Z_3} \quad (6)$$

$$\mathcal{L} = i Z_2 \bar{\psi} \cancel{D} \psi - Z_m Z_2 m \bar{\psi} \psi - Z_3 \frac{F^2}{4} - Z_1 e \bar{\psi} \cancel{A} \psi \quad (7)$$

$$Z_i = 1 + \delta_i \quad (8)$$

2. SCALING

In DimReg $e \rightarrow \mu^{(4-d)/2} e$ where $[\mu] = 1$.

$$\mathcal{L} = i Z_2 \bar{\psi} \cancel{D} \psi - Z_m Z_2 m \bar{\psi} \psi - Z_3 \frac{F^2}{4} - Z_1 \mu^{(4-d)/2} e \bar{\psi} \cancel{A} \psi \quad (9)$$

3. COUNTERTERMS

Schwartz gives the counterterms in the minimal subtraction scheme in DimReg (Eqs. 19.66 & 23.25).

$$\delta_1 = \delta_2 = -\frac{e^2}{8\pi^2\epsilon}, \quad \delta_3 = -\frac{e^2}{6\pi^2\epsilon}, \quad \delta_m = -\frac{3e^2}{8\pi^2\epsilon}, \quad \delta_e = \frac{e^2}{12\pi^2\epsilon}, \quad \epsilon \rightarrow 0 \quad (10)$$

4. RENORMALIZATION GROUP EQ.

The bare coupling is independent of μ , which implies

$$\mu \frac{de_B}{d\mu} = 0. \quad (11)$$

This is the renormalization group Eq. (RGE). Comparing Eqs. 1 & 9 yields

$$e_B \bar{\psi}_B \cancel{A}_B \psi_B = Z_1 \mu^{(4-d)/2} e \bar{\psi} \cancel{A} \psi. \quad (12)$$

$$Z_1 = Z_e Z_2 \sqrt{Z_3}, \quad \psi_B = \sqrt{Z_2} \psi, \quad \vec{A}_B = \sqrt{Z_3} \vec{A} \quad (13)$$

$$e_B \bar{\psi}_B \cancel{A}_B \psi_B = Z_e Z_2 \sqrt{Z_3} \mu^{(4-d)/2} e \bar{\psi} \cancel{A} \psi \quad (14)$$

$$e_B \bar{\psi}_B \cancel{A}_B \psi_B = Z_e \mu^{(4-d)/2} e \bar{\psi} \cancel{A} \psi \implies e_B = Z_e \mu^{(4-d)/2} e \quad (15)$$

The dimension d has ϵ dependence: $d = 4 - \epsilon$.

$$e_B = Z_e \mu^{\epsilon/2} e \quad (16)$$

I plug into Eq. 11.

$$0 = \mu \frac{d}{d\mu} (Z_e \mu^{\epsilon/2} e) \quad (17)$$

The renormalized coupling e depends on μ : $e = e(\mu)$.

$$0 = \mu \left(Z_e e \frac{d}{d\mu} \mu^{\epsilon/2} + \mu^{\epsilon/2} e \frac{dZ_e}{d\mu} + Z_e \mu^{\epsilon/2} \frac{de}{d\mu} \right) \quad (18)$$

I define the beta function:

$$\beta(e) = \mu \frac{de}{d\mu}. \quad (19)$$

$$0 = Z_e e \mu \frac{d}{d\mu} \mu^{\epsilon/2} + \mu^{\epsilon/2} e \mu \frac{dZ_e}{d\mu} + Z_e \mu^{\epsilon/2} \mu \frac{de}{d\mu} \quad (20)$$

$$0 = Z_e e \mu \frac{\epsilon}{2} \mu^{\epsilon/2-1} + \mu^{\epsilon/2} e \mu \frac{dZ_e}{d\mu} + Z_e \mu^{\epsilon/2} \beta(e) \quad (21)$$

$$0 = Z_e e \frac{\epsilon}{2} \mu^{\epsilon/2} + \mu^{\epsilon/2} e \mu \frac{dZ_e}{d\mu} + Z_e \mu^{\epsilon/2} \beta(e) \quad (22)$$

$$\begin{aligned} \frac{dZ_e}{d\mu} &= \frac{d}{d\mu} (1 + \delta_e) = \frac{d\delta_e}{d\mu} = \frac{d}{d\mu} \frac{e^2}{12\pi^2\epsilon} = \frac{1}{12\pi^2\epsilon} \frac{de^2}{d\mu} \\ &= \frac{e}{6\pi^2\epsilon} \frac{de}{d\mu} \end{aligned} \quad (23)$$

$$0 = \mu^{\epsilon/2} \left[Z_e e \frac{\epsilon}{2} + \frac{e^2}{6\pi^2\epsilon} \mu \frac{de}{d\mu} + Z_e \beta(e) \right] \quad (24)$$

$$0 = Z_e e \frac{\epsilon}{2} + \frac{e^2}{6\pi^2 \epsilon} \beta(e) + Z_e \beta(e) \quad (25)$$

$$0 = Z_e e \frac{\epsilon}{2} + \left(\frac{e^2}{6\pi^2 \epsilon} + Z_e \right) \beta(e) \quad (26)$$

$$0 = (1 + \delta_e) e \frac{\epsilon}{2} + \left(\frac{e^2}{6\pi^2 \epsilon} + 1 + \delta_e \right) \beta(e) \quad (27)$$

$$0 = \frac{\epsilon}{2} e + \frac{e^3}{24\pi^2} + \left(1 + \frac{e^2}{6\pi^2 \epsilon} + \frac{e^2}{12\pi^2 \epsilon} \right) \beta(e) \quad (28)$$

$$0 = \frac{\epsilon}{2} e + \frac{e^3}{24\pi^2} + \left(1 + \frac{e^2}{4\pi^2 \epsilon} \right) \beta(e) \quad (29)$$

$$\left(1 + \frac{e^2}{4\pi^2 \epsilon} \right) \beta(e) = -\frac{e}{2} \left(\epsilon + \frac{e^2}{12\pi^2} \right) \quad (30)$$

$$\beta(e) = -\frac{e}{2} \left(\epsilon + \frac{e^2}{12\pi^2} \right) \left(1 + \frac{e^2}{4\pi^2 \epsilon} \right)^{-1} \quad (31)$$

I expand for small e .

$$\boxed{\beta(e) = -\frac{\epsilon}{2} e + \frac{e^3}{12\pi^2}} \quad (32)$$

5. RUNNING COUPLING

I take $\epsilon \rightarrow 0$.

$$\beta(e) = \frac{e^3}{12\pi^2} \quad (33)$$

$$\mu \frac{de}{d\mu} = \beta(e) = \frac{e^3}{12\pi^2} \quad (34)$$

$$12\pi^2 \frac{de}{e^3} = \frac{d\mu}{\mu} \quad (35)$$

$$12\pi^2 \int_{e(\mu_0)}^{e(\mu)} \frac{de}{e^3} = \int_{\mu_0}^{\mu} \frac{d\mu}{\mu} \quad (36)$$

$$12\pi^2 \int_{e(\mu_0)}^{e(\mu)} \frac{dx}{x^3} = \int_{\mu_0}^{\mu} \frac{dx}{x} \quad (37)$$

$$6\pi^2 \left[\frac{1}{e(\mu_0)^2} - \frac{1}{e(\mu)^2} \right] = \ln \left(\frac{\mu}{\mu_0} \right) \quad (38)$$

$$\frac{1}{e(\mu_0)^2} - \frac{1}{e(\mu)^2} = \frac{1}{6\pi^2} \ln \left(\frac{\mu}{\mu_0} \right) \quad (39)$$

$$\frac{1}{e(\mu)^2} - \frac{1}{e(\mu_0)^2} = -\frac{1}{6\pi^2} \ln \left(\frac{\mu}{\mu_0} \right) \quad (40)$$

$$\frac{1}{e(\mu)^2} = \frac{1}{e(\mu_0)^2} - \frac{1}{6\pi^2} \ln \left(\frac{\mu}{\mu_0} \right) \quad (41)$$

$$\boxed{e(\mu)^2 = \frac{e(\mu_0)^2}{1 - \frac{e(\mu_0)^2}{6\pi^2} \ln \left(\frac{\mu}{\mu_0} \right)}} \quad (42)$$

Part II

Φ^3 in 6D

6. LAGRANGIAN

The bare Lagrangian is

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 + G_B \Phi_B - \frac{g_B}{3!} \Phi_B^3. \quad (43)$$

I renormalize.

$$\Phi_B = \sqrt{Z_\Phi} \Phi, \quad m_B = \sqrt{Z_m} m, \quad G_B = Z_G G, \quad g_B = Z_g g \quad (44)$$

$$\mathcal{L} = -Z_\Phi \frac{\Phi \square \Phi}{2} - Z_m Z_\Phi \frac{m^2}{2} \Phi^2 + Z_G \sqrt{Z_\Phi} G \Phi - Z_g Z_\Phi^{3/2} \frac{g}{3!} \Phi^3 \quad (45)$$

$$Z_1 = Z_G \sqrt{Z_\Phi}, \quad Z_2 = Z_g Z_\Phi^{3/2} \quad (46)$$

$$\mathcal{L} = -Z_\Phi \frac{\Phi \square \Phi}{2} - Z_m Z_\Phi \frac{m^2}{2} \Phi^2 + Z_1 G \Phi - Z_2 \frac{g}{3!} \Phi^3 \quad (47)$$

$$Z_i = 1 + \delta_i \quad (48)$$

7. DIMENSIONS & SCALING

In HW 4 I found

$$[\Phi] = \frac{d}{2} - 1. \quad (49)$$

$$[\mathcal{L}] = d = [g\Phi^3] = [g] + 3[\Phi] = [g] + \frac{3}{2}d - 3 \implies [g] = 3 - \frac{d}{2} \quad (50)$$

In $d = 6$: $[g] = 0$. To keep g dimensionless for all d I introduce the scale factor

$$g \rightarrow \mu^{3-d/2} g = \mu^{(6-d)/2} g, \quad [\mu] = 1. \quad (51)$$

$$\mathcal{L} = -Z_\Phi \frac{\Phi \square \Phi}{2} - Z_m Z_\Phi \frac{m^2}{2} \Phi^2 + Z_1 G \Phi - Z_2 \mu^{(6-d)/2} \frac{g}{3!} \Phi^3 \quad (52)$$

I used a different scaling in HW 4 so I need to be careful when copying results.

8. COUNTERTERMS

I found the counterterms in HW 4.

$$F = \frac{gm^2}{256\pi^3} \left[\gamma - \frac{3}{2} - \ln \left(4\pi \frac{\mu}{m^2} \right) \right] \quad (53)$$

Unfortunately this log is dimensional: I discuss this problem in HW 4. The Euler–Mascheroni constant is $\gamma \approx 0.577216$.

$$\delta_1 = -\frac{gm^2}{128\pi^3 F \epsilon}, \quad \delta_\Phi = -\frac{g^2}{384\pi^3 \epsilon}, \quad \delta_m = -\frac{5g^2}{384\pi^3 \epsilon}, \quad \delta_2 = -\frac{g^2}{64\pi^3 \epsilon} \quad (54)$$

$$Z_2 = Z_g Z_\Phi^{3/2} \implies \delta_g = \frac{1 + \delta_2}{(1 + \delta_\Phi)^{3/2}} - 1 \quad (55)$$

$$\delta_g = \left(1 - \frac{g^2}{64\pi^3 \epsilon} \right) \left(1 - \frac{g^2}{384\pi^3 \epsilon} \right)^{-3/2} - 1 \quad (56)$$

I expand to leading order as $g \rightarrow 0$.

$$\delta_g = \left(1 - \frac{g^2}{64\pi^3 \epsilon} \right) \left(1 + \frac{g^2}{256\pi^3 \epsilon} \right) - 1 \quad (57)$$

$$\delta_g = \frac{g^2}{256\pi^3 \epsilon} - \frac{g^2}{64\pi^3 \epsilon} \implies \boxed{\delta_g = -\frac{3g^2}{256\pi^3 \epsilon}} \quad (58)$$

9. RGE

Like the official solution I only consider the running of g (not G). This makes sense because the linear term in Eq. 52 is only there to kill tadpoles. The actual interaction is cubic.

$$\boxed{\beta(g) = \mu \frac{dg}{d\mu}}. \quad (59)$$

The RGE is

$$\mu \frac{dg_B}{d\mu} = 0. \quad (60)$$

Comparing Eqs. 43 & 52 yields

$$g_B \Phi_B^3 = Z_2 \mu^{(6-d)/2} g \Phi^3. \quad (61)$$

$$Z_2 = Z_g Z_\Phi^{3/2}, \quad \Phi_B = \sqrt{Z_\Phi} \Phi \quad (62)$$

$$g_B \Phi_B^3 = Z_g Z_\Phi^{3/2} \mu^{(6-d)/2} g \Phi^3 = Z_g \mu^{(6-d)/2} g \Phi_B^3 \implies g_B = Z_g \mu^{(6-d)/2} g \quad (63)$$

$$\mu \frac{dg_B}{d\mu} = \mu \frac{d}{d\mu} [Z_g \mu^{(6-d)/2} g] = 0, \quad d = 6 - \epsilon \quad (64)$$

$$\mu \frac{d}{d\mu} (Z_g \mu^{\epsilon/2} g) = 0 \quad (65)$$

$$\mu^{\epsilon/2} g \mu \frac{dZ_g}{d\mu} + Z_g g \mu \frac{d}{d\mu} \mu^{\epsilon/2} + Z_g \mu^{\epsilon/2} \mu \frac{dg}{d\mu} = 0 \quad (66)$$

$$\mu^{\epsilon/2} g \mu \frac{dZ_g}{d\mu} + \frac{\epsilon}{2} Z_g g \mu^{\epsilon/2-1} + Z_g \mu^{\epsilon/2} \beta(g) = 0 \quad (67)$$

$$\mu^{\epsilon/2} g \mu \frac{dZ_g}{d\mu} + \frac{\epsilon}{2} Z_g g \mu^{\epsilon/2} + Z_g \mu^{\epsilon/2} \beta(g) = 0 \quad (68)$$

$$\frac{dZ_g}{d\mu} = \frac{d}{d\mu} (1 + \delta_g) = \frac{d\delta_g}{d\mu} = -\frac{3}{256\pi^3\epsilon} \frac{dg^2}{d\mu} = -\frac{3g}{128\pi^3\epsilon} \frac{dg}{d\mu} \quad (69)$$

$$\frac{\epsilon}{2} Z_g g \mu^{\epsilon/2} + Z_g \mu^{\epsilon/2} \beta(g) - \frac{3g^2}{128\pi^3\epsilon} \mu^{\epsilon/2} \mu \frac{dg}{d\mu} = 0 \quad (70)$$

$$\mu^{\epsilon/2} \left[\frac{\epsilon}{2} Z_g g + Z_g \beta(g) - \frac{3g^2}{128\pi^3\epsilon} \beta(g) \right] = 0 \quad (71)$$

$$\frac{\epsilon}{2} Z_g g + Z_g \beta(g) - \frac{3g^2}{128\pi^3\epsilon} \beta(g) = 0 \quad (72)$$

$$\left(Z_g - \frac{3g^2}{128\pi^3\epsilon} \right) \beta(g) = -\frac{\epsilon}{2} Z_g g \quad (73)$$

$$\left(1 + \delta_g - \frac{3g^2}{128\pi^3\epsilon} \right) \beta(g) = -\frac{\epsilon}{2} (1 + \delta_g) g \quad (74)$$

$$\left(1 + \delta_g - \frac{3g^2}{128\pi^3\epsilon} \right) \beta(g) = -\frac{\epsilon}{2} g - \frac{\epsilon}{2} \delta_g g \quad (75)$$

$$\left(1 - \frac{9g^2}{256\pi^3\epsilon} \right) \beta(g) = -\frac{\epsilon}{2} g + \frac{3g^3}{512\pi^3} \quad (76)$$

$$\beta(g) = -\left(\frac{\epsilon}{2} g - \frac{3g^3}{512\pi^3} \right) \left(1 - \frac{9g^2}{256\pi^3\epsilon} \right)^{-1} \quad (77)$$

I expand as $g \rightarrow 0$.

$$\beta(g) = -\left(\frac{\epsilon}{2} g - \frac{3g^3}{512\pi^3} \right) \left(1 + \frac{9g^2}{256\pi^3\epsilon} \right) \quad (78)$$

$$\beta(g) = -\frac{\epsilon}{2} g \left(1 + \frac{9g^2}{256\pi^3\epsilon} \right) + \frac{3g^3}{512\pi^3} \quad (79)$$

$$\boxed{\beta(g) = -\frac{\epsilon}{2} g - \frac{3g^3}{256\pi^3}} \quad (80)$$

10. RUNNING COUPLING

I take $\epsilon \rightarrow 0$.

$$\beta(g) = -\frac{3g^3}{256\pi^3} \quad (81)$$

$$\mu \frac{dg}{d\mu} = -\frac{3g^3}{256\pi^3} \quad (82)$$

$$\frac{256\pi^3}{3} \frac{dg}{g^3} = -\frac{d\mu}{\mu} \quad (83)$$

$$\frac{256\pi^3}{3} \int_{g_0}^{g(\mu)} \frac{dx}{x^3} = -\int_{\mu_0}^{\mu} \frac{dx}{x}, \quad g_0 = g(\mu_0) \quad (84)$$

$$\frac{128\pi^3}{3} \left[\frac{1}{g_0^2} - \frac{1}{g(\mu)^2} \right] = -\ln \left(\frac{\mu}{\mu_0} \right) \quad (85)$$

$$\frac{1}{g(\mu)^2} - \frac{1}{g_0^2} = \frac{3}{128\pi^3} \ln \left(\frac{\mu}{\mu_0} \right) \quad (86)$$

$$\frac{1}{g(\mu)^2} = \frac{1}{g_0^2} + \frac{3}{128\pi^3} \ln \left(\frac{\mu}{\mu_0} \right) \quad (87)$$

$$\boxed{g(\mu)^2 = \frac{g_0^2}{1 + \frac{3g_0^2}{128\pi^3} \ln \left(\frac{\mu}{\mu_0} \right)}} \quad (88)$$